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Attacker

Purpose

The `attacker/` component contains controlled security-testing functionality for the OT/ICS testbed. Its role is to emulate an authorized attacker interacting with the industrial network and to validate how the PLC and SCADA environment responds to abnormal Modbus/TCP write operations.

Role in the Testbed

The repository describes the attacker container as a controlled Modbus write-test component. It is intended for isolated laboratory use rather than deployment against production systems.

Typical flow:

1. The attacker container joins the OT test network.
2. The target PLC exposes Modbus/TCP services.
3. The attacker sends a controlled Modbus request.
4. The PLC state changes according to its configured logic.
5. SCADA reflects the resulting process state.
6. Traffic can be captured and inspected with Wireshark.

Testing Objectives

- Validate that the PLC accepts expected Modbus/TCP requests.
- Understand Modbus function codes and packet fields.
- Observe the difference between normal traffic and controlled write activity.
- Verify that SCADA reflects PLC state changes.
- Provide repeatable traffic for defensive monitoring experiments.

Safety and Scope

Use this component only inside the isolated Docker-based OT laboratory. Tests should be deterministic, reversible, and documented before execution.

Related Components

- `openplc/` — simulated PLC and control logic
- `scadabr/` — SCADA/HMI
- `modbus-client/` — normal PLC read/write verification

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- `traffic/` — background normal Modbus traffic
- `scripts/` — environment and workflow automation

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